

Question Paper

Exam Date & Time: 06-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.Tech. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY
2025
SUBJECT: CSE 3122/CSE_3122 - SOFTWARE ENGINEERING

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Distinguish between generic and custom software products using suitable example. (2)
- 1B) List the short comings of exploratory software development process. Why development time and effort increase exponentially with linear increase in software size when exploratory development is used? (3)
- 1C) Implement a suitable process model and list activities in each phase for a travel company to develop and maintain its web and mobile applications to continuously enhance to adapt market changes. (5)
- 2A) Consider a software development company working on aircraft software development which involves high risks, complexity, safety and evolving requirements. As a project manager suggest a suitable process model for implementation and list activities in each phase. (5)
- 2B) Differentiate between inconsistent and incomplete requirements using suitable example? (2)
- 2C) Design and draw Decision tree and Decision table for library management software supporting new member, renewal, and cancel membership. (3)
- 3A) A software project involves designing a system for handling customer orders. The system has several modules, including order processing, payment handling, and inventory management. The design team has structured all modules in a single layer, allowing each module to call any other module directly. Evaluate the drawbacks of this design and suggest how layering could improve the system's structure. (2)
- 3B) Explain any 4 levels of cohesion and provide examples for each type. How does increasing cohesion improve the quality of a module? (3)
- 3C) Discuss the process of transforming a Data Flow Diagram (DFD) into a Structure Chart using Transform Analysis. Explain the steps involved in identifying the input, output, and central transform portions, and demonstrate how these portions are used to create the initial structure chart. (5)
- 4A) Illustrate, with an example, an Use Case Diagram. Show all it's parts. (5)
- 4B) How class diagram relationships are shown? Show their different symbols (3)
- 4C) Show Sequence Diagram with an example. (2)
- 5A) Illustrate Branch Coverage in White Box Testing with an example. (3)
- 5B) How do you perform Black Box Testing for finding square root of a number? Explain the steps involved. (5)
- 5C) How do you use Control Flow Graphs in White Box Testing? Explain with an example. (2)

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